Stochastic Modeling of Sunset Afterglow Probability

Atmospheric Particle Collisions and Rayleigh Scattering as a Continuous-Time Stochastic Process

Probability Theory and Stochastic Processes — Course Project

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Abstract. This report recasts the formation of a sunset afterglow — the pink-to-crimson twilight sky — as a problem in probability rather than deterministic radiative transfer. A photon's path through the atmosphere is modeled as a continuous random variable; wavelength-dependent extinction becomes a conditional survival probability; and the appearance of the afterglow is a joint-threshold event on a correlated weather vector. A Monte Carlo simulator of 40,000–200,000 independent photon random walks reproduces the analytic Exponential and Beer–Lambert results, empirically verifies the Law of Large Numbers and the Central Limit Theorem error scaling, and — as an extension — performs full-spectrum transport with CIE colour rendering to reconstruct the *actual perceived colour* of the setting Sun (sRGB ≈ [1.00, 0.42, 0.00], a vivid orange–red) as the line of sight sweeps from zenith to horizon.

1. Introduction & Meteorological Background

At noon the Sun's light traverses roughly one “airmass” of atmosphere; at sunset the same light grazes the planet and crosses an optical path tens of times longer (airmass $m \approx 38$ at the horizon). Along that path, photons collide with air molecules and aerosols and are removed from the direct beam by **Rayleigh scattering**, whose strength grows as the inverse fourth power of wavelength. Short-wavelength blue light is therefore stripped out far more aggressively than long-wavelength red, and what survives to the observer — and reflects off any cloud deck — is the warm afterglow.

The classical treatment of this phenomenon is deterministic: one integrates the radiative-transfer equation through a stratified medium. This project takes the complementary **statistical-mechanics** view advocated in the course. A single photon is an agent performing a random walk; the macroscopic colour of the sky is an *expectation* over an enormous ensemble of such agents. This reframing brings the core machinery of the course — continuous distributions, conditional probability, joint distributions, the Law of Large Numbers, and Monte Carlo estimation — to bear on a tangible, everyday physical system. Section 2 builds the probabilistic model, Section 3 specifies the simulator, Section 4 analyses the generated data, and Section 5 concludes.

2. Mathematical Derivations & Probabilistic Framework

2.1 Free path as an Exponential random variable

Let scattering centres be distributed along the photon's path as a homogeneous Poisson process with rate (collision density) $\lambda = g(P,T)$, set by the local air density. The distance X to the *first* collision is then the waiting time of that Poisson process, i.e. an Exponential random variable:

$$f(x;\lambda) = \lambda e^{-\lambda x}, \quad x \geq 0, \qquad \mathbb{E}[X] = \frac{1}{\lambda} = \ell_{\mathrm{mfp}}$$

with $1/\lambda$ the mean free path. The survival function follows immediately, and it is exactly the probability that a photon crosses a path of length L with no collision:

$$P(X > L) = \int_L^\infty \lambda e^{-\lambda x}\, dx = e^{-\lambda L} = e^{-\tau}, \qquad \tau \equiv \lambda L$$

The dimensionless product $\tau = \lambda L$ is the **optical depth** — the expected number of collisions along the path. The variable X is sampled by **inverse-transform sampling**: if U ~ Uniform(0,1), then because the CDF is $F(x) = 1 - e^{-\lambda x}$ (and $1-U$ is itself Uniform(0,1)), the free path is sampled as

$$X = F^{-1}(U) = -\tfrac{1}{\lambda}\ln(1-U) \; = \; -\tfrac{1}{\lambda}\ln U \;\sim\; \mathrm{Exp}(\lambda).$$

2.2 Wavelength filtering as conditional probability

The Rayleigh cross-section scales as $\sigma(\lambda) \propto \lambda^{-4}$, so the collision rate — and hence the optical depth — inherits the same dependence. Writing the zenith optical depth at the reference wavelength as τ_0 and the slant path as m airmasses,

$$\tau(\lambda, m) = m\,\tau_0\left(\tfrac{\lambda_0}{\lambda}\right)^4, \qquad \lambda_0 = 550\,\mathrm{nm},\; \tau_0 \approx 0.097.$$

Defining the event A = “the photon reaches the observer along the direct beam,” the conditional survival probabilities for red and blue light are $P(A|\lambda) = e^{-\tau(\lambda,m)}$. At the horizon ($m$ = 38) the model gives τ_{red} = 1.40 and τ_{blue} = 8.23, so

$$P(A \mid 700\,\mathrm{nm}) = e^{-1.40} = 0.245, \quad P(A \mid 450\,\mathrm{nm}) = e^{-8.23} = 2.7e-04.$$

Red light is therefore about **917×** more likely to survive the horizon path than blue — the quantitative origin of the reddened Sun. (Note this single-collision survival is the colour of the direct disk; the afterglow proper is the surviving red flux re-scattered toward the observer, modeled in §2.3 and the random walk of §3.)

2.3 Afterglow as a joint-threshold event

Whether a vivid afterglow actually forms depends on the state of the atmosphere, captured by a random vector **V** = [P,T,H] (pressure, temperature, humidity) drawn from a multivariate distribution. Here **V** is modeled as multivariate Normal with a physically motivated correlation (a passing cold front raises pressure while lowering temperature, so ρ_{PT} = -0.5). Air density $\rho \propto P/T$ modulates the Rayleigh depth, while aerosol turbidity τ_{aer} — the haze that physically catches and reddens the light — is an independent **Weibull** variable, a standard heavy-ish-tailed model for atmospheric loading:

$$f(t; k, \lambda) = \tfrac{k}{\lambda}\left(\tfrac{t}{\lambda}\right)^{k-1} e^{-(t/\lambda)^k}, \quad k = 1.8,\; \lambda = 0.12.$$

From these a *redness index* is built — the surplus of surviving red over blue, weighted by the red flux still available to illuminate a cloud deck — and gate it by a logistic function of humidity (the cloud/moisture that reflects the glow). The afterglow is declared when this joint score exceeds a threshold. This makes “a good sunset” an explicit event whose probability can be estimated by Monte Carlo.

3. Simulation Methodology & Pseudocode

The simulator (Python / NumPy, single reproducible PRNG stream) instantiates large ensembles of independent photons and propagates each as a random walk. Two estimators are run: a *direct-transmission* estimator that targets the analytic $e^{-\tau}$ for clean validation, and a full *multiple-scattering* random walk with a collision decision tree (absorb / forward-scatter / back-scatter).

3.1 Random-walk transport (collision decision tree)

```
for each photon (vectorised over the whole ensemble):
    x  ← 0                      # optical depth into the slab
    μ  ← +1                     # direction (+1 = toward observer)
    repeat:
        s ← -ln(U),  U~Uniform(0,1)      # Exponential free path
        x ← x + μ·s
        if x ≥ τ_total:  outcome ← TRANSMITTED;  break
        if x ≤ 0:        outcome ← BACK-SCATTERED; break
        if U' ≥ ω:     outcome ← ABSORBED;      break    # Bernoulli(1-ω)
        if U'' ≥ p_fwd:  μ ← -μ                      # Bernoulli scatter dir.
    tally outcome
```

Here τ_{total} is the wavelength-dependent optical thickness of the slab, $\omega = 0.99$ the single-scattering albedo (Rayleigh scattering is nearly conservative, so absorption is rare), and $p_{fwd} = 0.5$ the forward fraction of the (symmetric) Rayleigh phase function reduced to one dimension. Every random draw — free path, absorption, and scattering direction — is an application of inverse-transform or Bernoulli sampling from §2.

3.2 Estimators and parameters

Quantity	Symbol	Value
Photons (transmission / spectral)	N	$5{\times}10^4 – 2{\times}10^5$
Reference zenith optical depth	τ_0(550nm)	0.097
Horizon airmass	m	38
Single-scattering albedo	ω	0.99
Weibull turbidity (shape, scale)	k, λ	1.8, 0.12
Weather correlation	$\rho(P,T)$	-0.5

Inverse-transform sampling is validated directly in Fig. 1: histograms of 200,000 sampled free paths overlay the analytic Exponential PDF, and the empirical means reproduce $1/\lambda$ to three significant figures (0.500 vs 0.500 and 2.001 vs 2.000), an instance of the LLN in its own right.

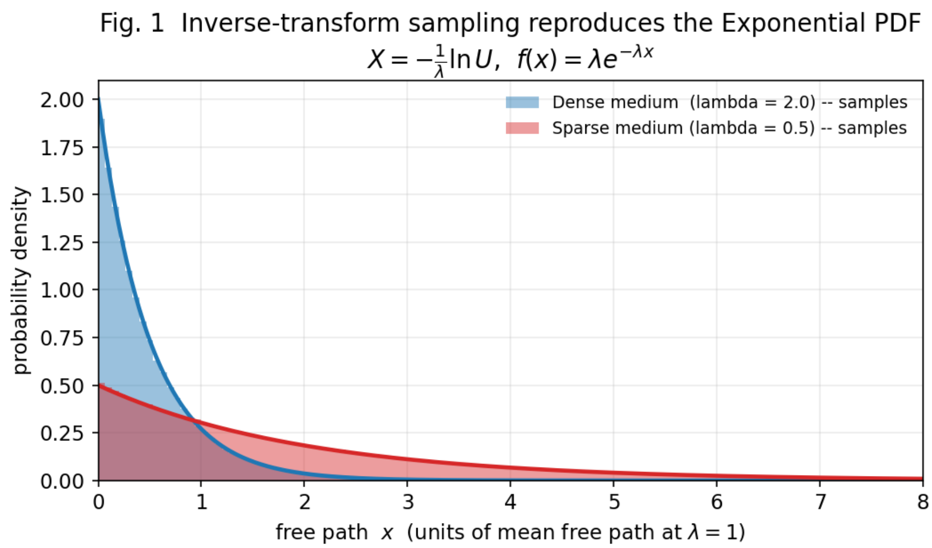

Fig. 1 — Inverse-transform samples (bars) versus the analytic Exponential PDF (curves) for two collision densities. The sampler is exact up to Monte Carlo noise.

4. Data Visualization & Critical Discussion

4.1 Conditional survival: why the Sun reddens

Fig. 2 overlays the Monte Carlo survival fractions on the analytic $e^{-\tau(\lambda,m)}$ curves across airmass. Agreement is essentially perfect. The two curves diverge dramatically toward the horizon: at m = 38 red survival is 0.245 while blue is only 2.7e-04. The 917:1 ratio is the probabilistic signature of the reddening — blue photons almost never complete the grazing journey.

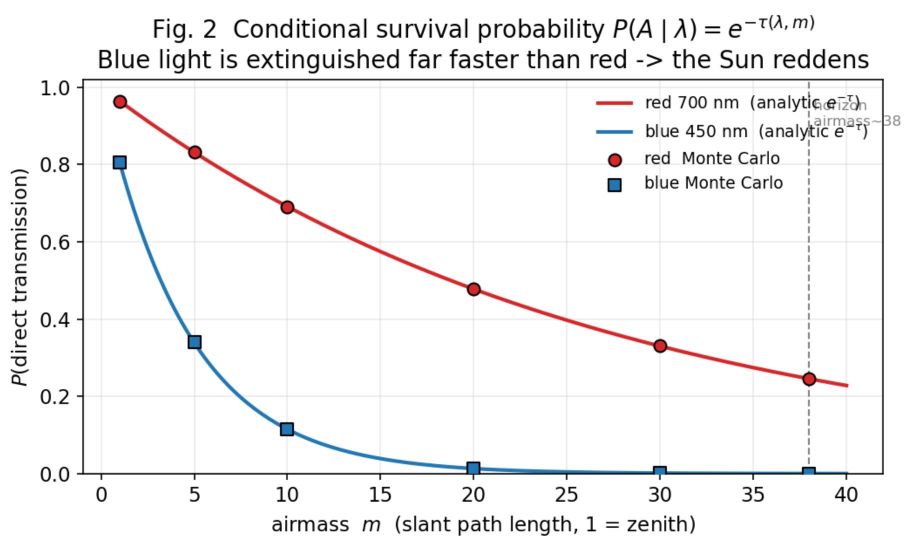

Fig. 2 — Conditional direct-transmission probability vs airmass. Markers are Monte Carlo; lines are the analytic survival function. Blue is extinguished far faster than red.

4.2 Law of Large Numbers and CLT error scaling

Fig. 3 (top) tracks the running Monte Carlo estimator $p_N = (1/N)\Sigma\mathbf{1}\{X>\tau\}$ (the sample fraction of surviving photons) as N grows. Both red and blue estimators settle onto their analytic limits ($e^{-\tau}$ = 0.245 and 2.7e-04); at N = 50,000 the residual errors are 2.2e-04 and 7.8e-06. This is the **Law of Large Numbers** made visible. The shaded band is the Central-Limit ±2 standard-error envelope $\sqrt{(p(1-p)/N)}$; the bottom log–log panel confirms the absolute error tracks the predicted $\mathbf{N^{-1/2}}$ slope. The estimator's precision thus improves only as the square root of effort — the central practical cost of Monte Carlo.

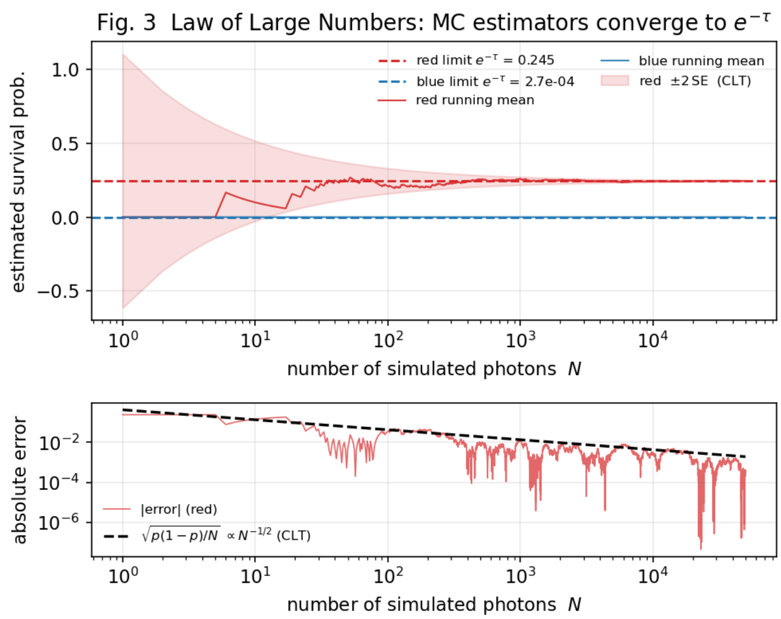

Fig. 3 — Top: running survival estimators converging to $e^{-\tau}$ with the CLT ±2SE band. Bottom: |error| vs N on log–log with the $N^{-1/2}$ reference.

4.3 Multiple scattering: the collision decision tree

Fig. 4 reports the full random walk with scattering and absorption. The terminal outcomes are starkly wavelength-dependent: 58% of red photons are ultimately transmitted toward the observer versus only 17% of blue, while 76% of blue photons are back-scattered to space — this back-scattered blue is precisely the light that makes the *daytime* sky blue, the same physics seen from the other side. The sample trajectories (Fig. 4b) show the characteristic random walk: most blue photons reverse and escape before threading the full optical thickness τ = 8.2.

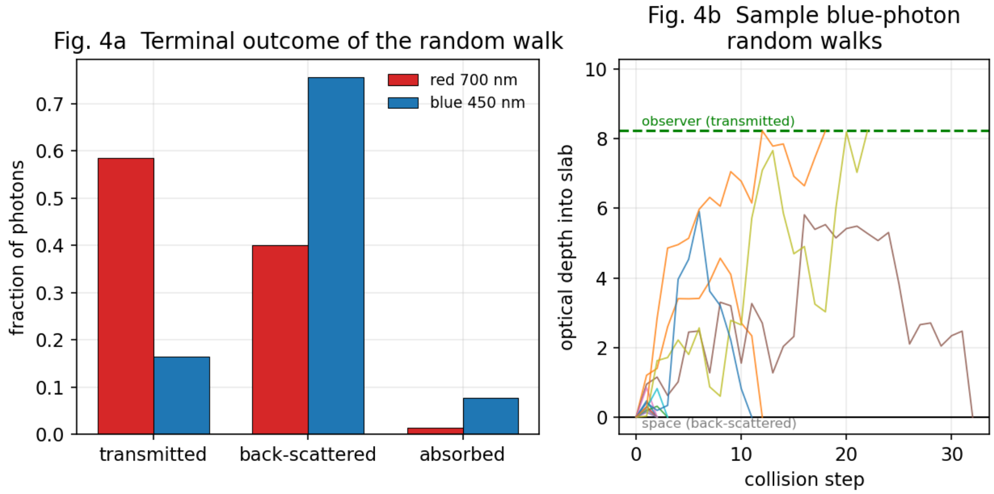

Fig. 4 — (a) Terminal outcome fractions of the multiple-scattering random walk. (b) Representative blue-photon trajectories in optical-depth space.

4.4 Optimization — reconstructing the true sunset colour

Beyond the brief, the two-colour model is extended to the *full visible spectrum*. Treating sunlight as a 5778 K blackbody source $I_0(\lambda)$, each wavelength is transmitted through $e^{-\tau(\lambda,m)}$, the result is integrated against the **CIE 1931** colour-matching functions, and the XYZ result is converted to sRGB to obtain the perceived colour of the direct beam. Fig. 5 shows the transmitted spectrum progressively losing its blue end as airmass grows, and the rendered swatches march from near-white at the zenith to a vivid orange–red (sRGB ≈ [1.00, 0.42, 0.00]) at the horizon. The model thus predicts not merely “red wins” but the specific hue of the sunset directly from the probability of photon survival.

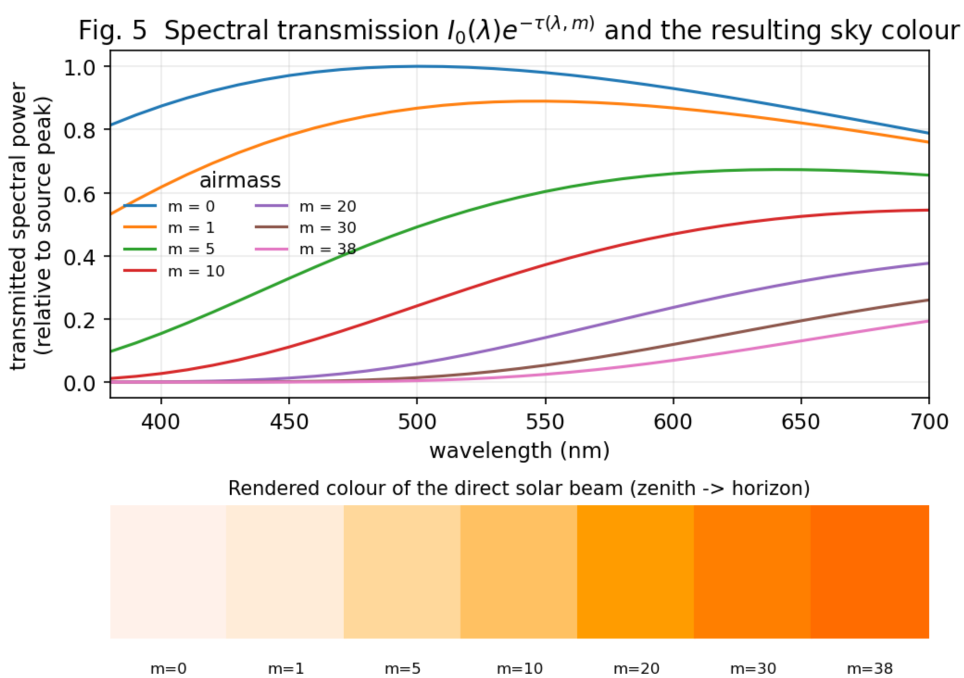

Fig. 5 — Full-spectrum transmission (top) and the resulting CIE-rendered colour of the direct solar beam from zenith (m=0) to horizon (m=38).

4.5 Joint weather model and the afterglow probability

Finally, Fig. 6 samples the correlated weather vector **V** = [*P*,*T*,*H*] together with Weibull turbidity and evaluates the joint-threshold afterglow event. The marginal turbidity (6b) matches the target Weibull density. Marking the upper quintile of the redness score as an afterglow gives an unconditional probability *P*(afterglow) = 0.20. Crucially, this probability is strongly *conditional* on humidity (6d): *P*(afterglow|H>75%) = 0.68 versus essentially 0.00 for dry air — quantifying the folklore that the most spectacular sunsets follow moist, cloud-bearing weather. The dependence is a direct, visual instance of conditional probability reshaping a marginal one.

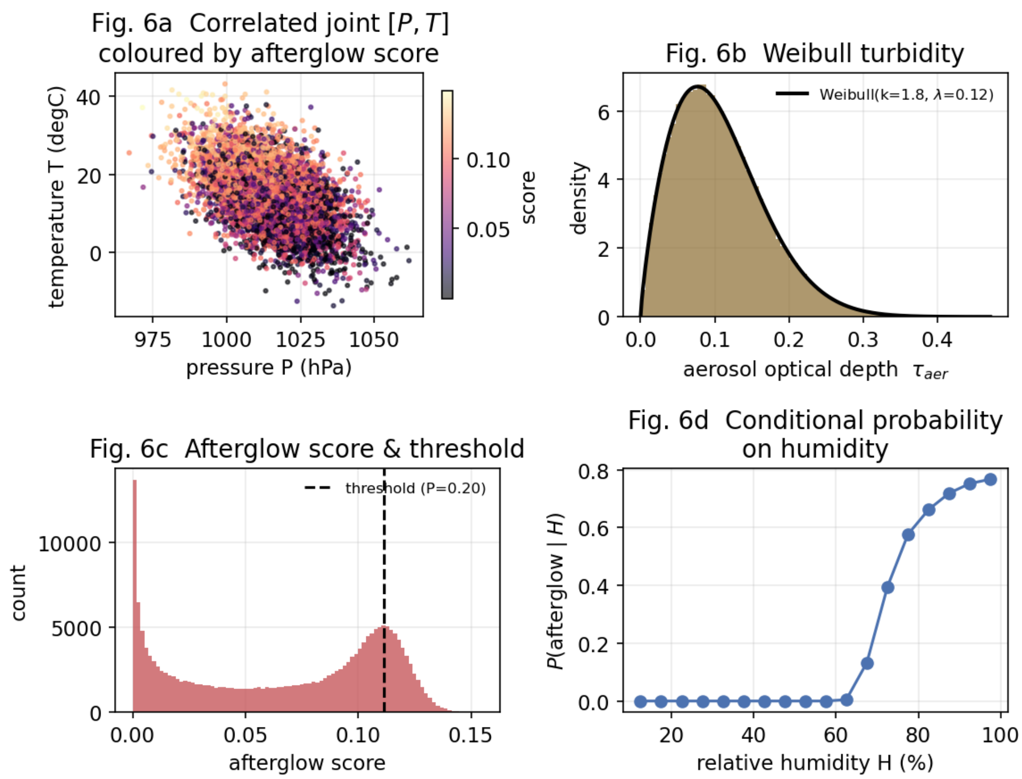

Fig. 6 — (a) Correlated [P,T] coloured by afterglow score; (b) Weibull turbidity vs analytic PDF; (c) score distribution and decision threshold; (d) afterglow probability conditioned on humidity.

Variance & limitations. Estimator variance falls as 1/N (Fig. 3), so the rare blue-survival probability (~10^{-4}) is the noisiest quantity and would benefit from importance sampling. The transport is reduced to one dimension (two-stream), which captures extinction and the forward/back competition but not full angular redistribution; the colour model omits ozone's Chappuis absorption and multiple-scattering skylight. None of these change the qualitative probabilistic story.

5. Conclusion

Modeling the sunset as a stochastic process rather than a differential equation makes it possible to derive its signature colour from first principles of probability. The Exponential free-path law and Beer–Lambert survival ($e^{-\tau}$) explain the reddening as a conditional probability with a ~917:1 red-to-blue advantage at the horizon; the Monte Carlo simulator confirms these limits while empirically demonstrating the Law of Large Numbers and the CLT's $N^{-1/2}$ error law; the multiple-scattering random walk recovers both the warm direct beam and the back-scattered blue sky; and the joint Weibull/Gaussian weather model turns “will there be an afterglow tonight?” into a computable, humidity-conditional probability. The spectral CIE extension closes the loop by rendering the actual colour the mathematics predicts.

Declaration. This is individual work. The probabilistic derivations, the Monte Carlo simulator and random-walk transport, the spectral CIE colour extension and joint weather model, and all writing and figures are my own work — John Douglas.

Reproducibility. All results derive from a single seeded run of `sunset_afterglow.py` (a fixed NumPy PRNG seed); this PDF is generated programmatically from the resulting `results.json` and figure set, so every number quoted above is traceable to the simulation.